

Aarya Arban

T11 - 05

Assignment No. 5

AIM: Implement any 5 converters like currency converter, temperature converter, distance converter, time converter, weight converter/speed converter in Angular js

LO3 – Implement Single page applications using Angular JS

Theory:

 [AngularJS Overview](#)

AngularJS is a **JavaScript-based open-source front-end web framework** mainly used to build **dynamic single-page applications (SPAs)**. It was developed by **Google** and first released in **2010**.

-
-  **Key Features of AngularJS**
1. **MVC Architecture** o AngularJS uses the **Model-View-Controller** pattern to separate application logic, UI, and data.
 - o This helps in organizing code and making it more manageable.
 2. **Two-Way Data Binding** o Any change in the model automatically updates the view and vice versa.
 - o It reduces the amount of boilerplate code needed to keep the view and model in sync.
 3. **Directives** o Custom HTML attributes like ng-model, ng-bind, ng-repeat, etc., which extend the functionality of HTML. o Developers can also create **custom directives**.
 4. **Dependency Injection (DI)** o AngularJS has a built-in **DI system** that helps manage and inject dependencies like services or factories efficiently.

5. **Templates** o Uses regular HTML that is enhanced with AngularJS directives to create dynamic views.
6. **Routing** o ngRoute module allows navigation between different views of the application, turning it into a **Single Page Application**.
7. **Filters** o Used to format the data displayed to the user (e.g., currency, date, uppercase/lowercase).
8. **Services & Factories** o AngularJS provides services (like \$http) that handle specific tasks such as making AJAX calls.
 - o You can also create your own reusable services.

PROGRAM:

INDEX.HTML

```
<!DOCTYPE html>
```

```
<html lang="en" ng-app="converterApp">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Converter App</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
  <script src="https://code.angularjs.org/1.8.2/angular.min.js"></script>
```

```
  <style>    body {
```

```
fontfamily: Arial, sans-serif;
```

```
margin: 0;      padding: 0;
```

```
background-color: #f4f4f4;
```

```
    }

</style>

</head>

<body>

<div class="container" ng-controller="ConverterController">

    <h2>Conversion Tools</h2>

    <!-- Currency Converter -->

    <div class="converter-box">

        <h3>Currency Converter: USD to INR</h3>

        <input type="number" ng-model="currencyAmount" placeholder="Enter
Amount in USD" />

        <button ng-click="convertCurrency()">Convert</button>

        <div class="result" ng-if="convertedCurrency !== undefined">

            {{currencyAmount}} USD = {{convertedCurrency}} INR

        </div>

    </div>

</div>

    <!-- Temperature Converter -->

    <div class="converter-box">

        <h3>Temperature Converter: Celsius to Fahrenheit</h3>

        <input type="number" ng-model="tempCelsius" placeholder="Enter
Temperature in Celsius" />
```

```
<button ng-click="convertTemperature()">Convert</button>
```

```
<div class="result" ng-if="convertedTemp !== undefined">
```

```
    {{tempCelsius}} °C = {{convertedTemp}} °F
```

```
</div>
```

```
</div>
```

```
<!-- Distance Converter -->
```

```
<div class="converter-box">
```

```
    <h3>Distance Converter: Kilometers to Miles</h3>
```

```
    <input type="number" ng-model="distanceKm" placeholder="Enter Distance  
in Kilometers" />
```

```
    <button ng-click="convertDistance()">Convert</button>
```

```
    <div class="result" ng-if="convertedDistance !== undefined">
```

```
        {{distanceKm}} Km = {{convertedDistance}} Miles
```

```
    </div>
```

```
</div>
```

```
<!-- Time Converter -->
```

```
<div class="converter-box">
```

```
    <h3>Time Converter: Hours to Minutes</h3>
```

```
    <input type="number" ng-model="timeHours" placeholder="Enter Time in  
Hours" />
```

```
    <button ng-click="convertTime()">Convert</button>
```

```

    <div class="result" ng-if="convertedTime !== undefined">

        {{timeHours}} Hour(s) = {{convertedTime}} Minutes

    </div>

</div>

<!-- Weight Converter -->

<div class="converter-box">

    <h3>Weight Converter: Kilograms to Pounds</h3>

    <input type="number" ng-model="weightKg" placeholder="Enter Weight in
Kilograms" />

    <button ng-click="convertWeight()">Convert</button>

    <div class="result" ng-if="convertedWeight !== undefined">

        {{weightKg}} Kg = {{convertedWeight}} lbs

    </div>

</div>

</div> <script>    var app =
angular.module('converterApp', []);
app.controller('ConverterController',
function($scope) {

```

```
// Currency conversion (example: 1 USD = 82
INR)    $scope.convertCurrency = function() {
if ($scope.currencyAmount) {

    $scope.convertedCurrency = $scope.currencyAmount * 87;

}

};
```

```
// Temperature conversion (Celsius to Fahrenheit)
$scope.convertTemperature = function() {    if
($scope.tempCelsius) {

    $scope.convertedTemp = ($scope.tempCelsius * 9/5) + 32;

}

};
```

```
// Distance conversion (Kilometers to Miles)
$scope.convertDistance = function() {    if
($scope.distanceKm)    {

    $scope.convertedDistance = $scope.distanceKm *
0.621371;

}

};
```

```
// Time conversion (Hours to
Minutes)    $scope.convertTime =
function() {    if
($scope.timeHours) {
    $scope.convertedTime = $scope.timeHours * 60;
    }
};

// Weight conversion (Kilograms to Pounds)
$scope.convertWeight = function() {    if
($scope.weightKg) {
    $scope.convertedWeight = $scope.weightKg * 2.20462;
    } };
});

</script>

</body>

</html>
```

STYLE.CSS

```
.container { width: 70%; margin: 50px  
auto; background-color: #fff; padding:  
30px; box-shadow: 0 4px 8px rgba(0, 0,  
0, 0.1);
```

```
} h2 { text-align:  
center; }
```

```
.converterbox {  
margin:  
20px 0; }
```

```
.converter-box input,  
.converter-box select {  
width: 100%; padding:  
10px; margin: 10px 0;  
border: 1px solid #ccc;  
border-radius: 4px;  
}
```

```
.converter-box button {  
padding: 10px 20px;
```



```
background-color: rgb(110,
196, 70); color: white;
border: none; border-radius:
4px; cursor: pointer;
}
.converter-box button:hover {
backgroundcolor: #0056b3; } .result {
textalign: center;

font-size: 1.2em;

color: #333;
}
```

OUTPUT:

Conversion Tools

Currency Converter: USD to INR

Convert

1 USD = 87 INR

Temperature Converter: Celsius to Fahrenheit

Convert

100 °C = 212 °F

Distance Converter: Kilometers to Miles

Convert

1000 Km = 621.371 Miles

Time Converter: Hours to Minutes

Convert

1 Hour(s) = 60 Minutes

Weight Converter: Kilograms to Pounds

Convert

10 Kg = 22.0462 lbs